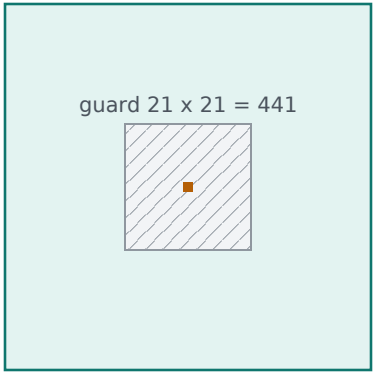
a The ring around one pixel



ring = 3,721 - 441
= 3,280 pixels

score = |X - ring mean|^2

fewer than 60 valid in the ring (2 x 30): global-mean score

b A box sum from four look-ups

values

1	2	0	1
3	1	2	0
0	2	1	3
1	0	2	1

integral image (zero-led)

0	0	0	0	0
0	**1**	3	**3**	4
0	4	7	9	10
0	**4**	9	**12**	16
0	5	10	15	20

shaded 2 x 2 box: 12 - 3 - 4 + 1 = 6

check: 1 + 2 + 2 + 1 = 6

Four look-ups whatever the box size: 61 x 61 costs the same as 2 x 2. The ring = outer box - guard box.